Reading Tabular KDB Data

This section uses an excerpt of khipu data to introduce the standard conventions of khipu recording, as first practiced by the Aschers (see note 4). Figure 1 shows the data for cords 28 through 37 (including cord 37's subsidiary cord) of MM002, which is in the Landesmuseum Detmold. MM002 was recorded from left to right by the author; as such, cord 28 is the twenty-eighth cord from the left, viewing the khipu as mounted. The data excerpt is reproduced below:

	A	B	C	D	E	F	G	I	J	K	L
1											
2	**KDB data as of 12/6/2018 9:30 AM EST**										
3											
4	**KHIPU MM002**										
5											
6	Cord Nun	Spir	Ply	Attch	Knots	Length	Term	Color	Value	Alt. Values	Subs Pos
34	28	Z	S	V	3S(14.5,Z)	37.0	K	MB	30		
35	29	Z	S	V	2L(29.5,Z)[U]	38.0	K	W:YB	2		
36	30	Z	S	V	5S(13,S); 4S([21.5,S],[23.5,Z],[25,S],[26,Z])	30.0	K	YB	54	540	
37	31	Z	S	V	9L(28.5,Z)[U]	32.0	K	AB	9		(1:14.5)
38	31s1	Z	S		4L(15.5,Z)[U]	21.5	R	AB:YB	4		
39	32	Z	S	V	8L(28,Z)[U]	35.0	K	YB	8		(1:18.5)
40	32s1	Z	S		6L(9,Z)[U]	14.5	K	AB:YB	6		
41	33	Z	S	?		25.0	B/R	AB	0		
42	34	Z	S	V	8S(8.5,S)	36.5	K	EB	80		
43	35	Z	S	V	1S(6.5,S);3S(12,S);2S(18,S);5L(25,S)[U]	31.0	B	MB:KB	1325		
44	36	Z	S	V	1S(7.5,Z);2S(12.5,Z)	40.5	K	YB	120		(1:14.5)
45	36s1	Z	S		3S(3.5,S);5L(16,S)[U]	27.0	R	MB	35		
46	37	Z	S	V	1S(8,S);5L(35.5,Z)[U]	42.0	K	YB	15		(1:19.5)
47	37s1	Z	S		3S(3,S);5L(10.5,Z)[U]	15.0	K	LB	35		

Figure 1: Excerpt of MM002 Data

Eleven columns are included in the figure. A twelfth column (not included above) captures notes on unusual knots, color patterns, damage to the artifact or other characteristics which merit special attention in subsequent research. The Cord Number column (column "A" in the above) records the ordered label for the specified cord. As introduced above, cord number 12 refers to the twelfth cord of the khipu, as read from left to right. Subsidiary cords (referring to khipu cords which are attached to pendant cords, rather than to the primary horizontal parent string) are denoted by a lowercase "s"—that is, the subsidiary cord of pendant cord number 31 is entered as cord number 31s1 in figure 1. The Spin and Ply columns ("B" and "C" in the figure) capture the construction of the cord itself. While spin refers to the direction of the fibers constituting the cord, ply captures the overall twisting of the cord. Both categories are binary: fibers either follow the axial "Z" or axial "S." That is, the dominant twist of an S-ply cord, when viewed closely, proceeds from high and left to low and right.

"Attch" refers to attachment knots (column "D" in the figure). An extended quotation from my recent work helps to introduce this concept:

> During khipu construction, each pendant cord is attached to the primary cord with what is designated as an attachment knot. This maneuver presents the khipu creator, or khipukamayuq, with two choices: one can loop the final tie from below, then up and around the primary cord toward the body, forming a verso knot; or one can perform the operation by moving the cord up and over the primary cord, away from the body, forming a recto knot…
>
> It is important to bear in mind that the directional reading order for noting the recto/verso distinction along the primary cords… follows the standard recording/reading procedure practiced by students of the khipus, whereby reading commences at the "head"—the knotted or tasseled end of the primary cord—and proceeds from left to right, terminating at the primary cord's dangling "tail."[7]

Despite the convention described in the excerpt (i.e. reading from left to right, from head to tail), khipus studied often lack this contextual information. For example, MM012 (one of the ten khipus sewn onto the black fabric backing of IVc.366.03 in Basel) is fragmentary, with attached pendant cords tightly and evenly spaced along the entirety of the primary cord. In this case, I defaulted to recording from left to right. It is worth noting that the recorded knot directions are easily reversed algorithmically, if further information arises that implies a different reading direction. Nevertheless, this fact is often relevant only among khipu researchers. Attachment knots are depicted below in figure 2:

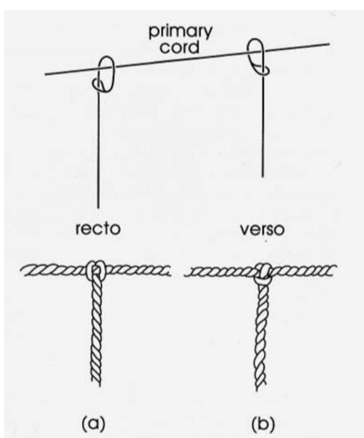

Figure 2: Attachment Knot Directions[8]

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[7] Manuel Medrano and Gary Urton, "Toward the Decipherment of a Set of Mid-Colonial Khipus from the Santa Valley, Coastal Peru," Ethnohistory 65, no. 1 (2018): 6-7.
[8] Ibid., 6.

Attachment knots are thus entered using a binary R/V designation (recto/verso). It is possible that an attachment direction is ambiguous, most often due to cord damage and deterioration. A question mark "?" replaces the R/V designation in such cases. Subsidiary cords do not have associated attachment knots by convention, as the attachment is defined relative to the primary cord. (However, subsidiaries are attached directly to pendant cords.)

Cord knots are recorded in column "E" in figure 1. Each row represents one khipu cord. The thirteen khipus studied during my research trip (with the possible exception of MM017, described below) are numerical in nature. That is, each cord records numerical values using a base-10 hierarchical knot system, whereby the reader proceeds in magnitude up the pendant cord, from units to tens, hundreds, thousands, etc.—such that the largest numerical values are found nearest to the horizontal primary cord. Three types of knots (with numerical values) are recognized in standard khipus: single knots (tens, hundreds, thousands, etc.), long knots (2-9) and figure-eight knots (1).[9] For example, take the entry in cell E43, which records the knots of pendant cord 35. The entry is as follows: "1S(6.5,S);3S(12,S);2S(18,S);5L(25,S)[U]." This sequence is intimidating upon initial observation. Nevertheless, the interpretation is simple. One single knot appears along this pendant, 6.5cm from the primary cord, tied in an "S" direction (referring to the axial direction of tying introduced above). Three single knots appear 12.0cm from the primary cord, each tied in "S" directions. Two additional single knots are tied 18.0cm from the primary cord, again in the "S" direction. Finally, a long knot with five turns (signing the number "5") is tied 25.0cm from the primary cord, along an "S" radial axis and with the straight edge of the pendant cord emanating up "[U]" from the knot, as viewed face-on (as opposed to down "[D]"). While not included in this example, an "E" denotes a figure-eight knot—recognized as a "1" on all numerical khipus.

The remaining columns complete the description of each cord. Length (column "F") measures each cord in cm from its attachment to the end of its fibers. Note that this measurement operates differently in the cases of pendant and subsidiary cords. For pendant cords, the cord is measured beginning from its attachment to the primary cord. For subsidiary cords, this measurement instead begins at the point of attachment to the pendant cord which hosts it. In

[9] See Ascher and Ascher [1981] 1997 for a broader introduction to khipu pendant knots.

other words, if a pendant and subsidiary cord are each recorded as 25.0cm in length, the reader would find the two cords to be the same length if placed next to each other. "Term" (column "G") measures the termination of the specified pendant/subsidiary cord. Options include knotted ("K"), raveled ("R") and broken ("B"). In some cases, two designations are used. For example, "K&R" or "K/R" describes a pendant cord which has both a terminating knot and associated cord material that dangles (raveled) from the conclusion of the knot. Color (column "I") is recorded using a variety of notational conventions. A range of colors are mapped to various abbreviations, following the convention first introduced by Carrie Brezine.

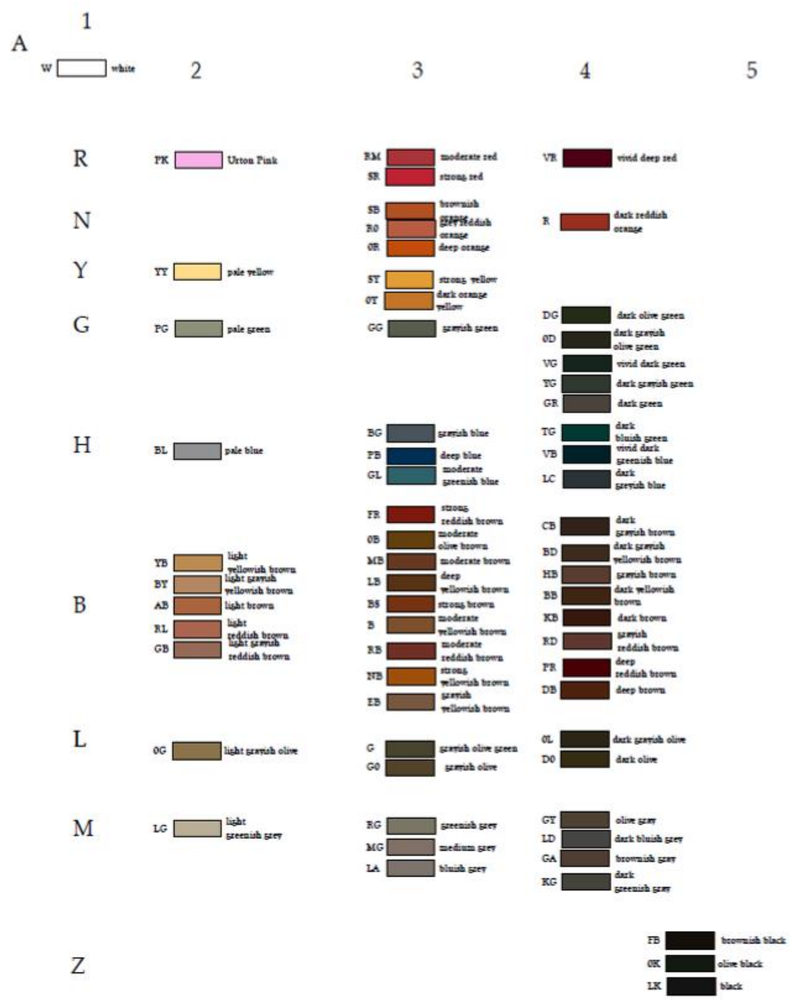

Figure 3: Brezine's Khipu Cord Color Matrix

Thus, an "AB" cord is light brown; a "KB" cord is dark brown, etc. Mottled cords (formed by interweaving two colors of constituent threads) are separated by a colon. Therefore, an "AB:KB" cord is mottled, combining light brown and dark brown—producing a speckled, uneven brown patterning. A barber pole designation is used to refer to cords that combine two or more colors in an alternating twist pattern. These cords are encoded with dashes, such that an “AB-KB” cord includes a symmetric alternation of light brown and dark brown—replicating the familiar patterning of a barber shop pole. Rarer cases include spliced or dyed cords, in which a cord’s color changes suddenly. Such cords are entered using a forward slash and associated measurement units: “AB (0-16.5) / KB (16.5-30),” for example, describes a single cord of length 30.0cm, which is solid light brown for 16.5cm, before changing colors abruptly to dark brown for the remaining duration of its length. The value column (“J”) maps the knots described in column E to the associated numerical values. Taking the above-introduced example of pendant cord 35: one single knot in the thousands place, three single knots in the hundreds place, two single knots in the tens place and a long knot of size five in the units place combine to form 1,325. In the case of cords broken close to their attachments, a question mark “?” is placed in the value column to capture uncertainty. A broken cord measuring 3.0cm could have recorded zero, 154 or 2,596—we lack the information to make a more determinate estimate.

In uncommon cases, the value of a cord remains ambiguous after recording its knots. In this situation, alternative values are recorded (column “K”) to capture alternate hypotheses considered by the researcher. Values entered in column J are believed to be more likely than those in column K, when both are included. Column J is never left empty: if column K includes an alternate value, then a primary value must reside in column J. Cord information concludes with column “L,” which records the number and location of subsidiary cords. For example, the entry in cell L39 reads “(1:18.5).” This notation implies that the referenced pendant cord has a single subsidiary cord, which is attached to the pendant cord 18.5cm down its length. This entry further implies that the subsequent row will record the subsidiary cord. In other words, a row for cord 97 which has an entry in column L implies that the subsequent row of data will describe cord 97s1—the subsidiary of cord 97. Alternatively, an entry of “(2:18.5)” would imply that the referenced cord has two subsidiaries, attached in direct succession 18.5cm down its length. The next *two* rows would in turn be occupied by the data from the two subsidiaries. Lastly, subsidiary cords themselves may have associated subsidiary cords, producing a branching appearance. This is recorded using multiple lowercase “s” in succession. For example, the first subsidiary *of the second subsidiary* to pendant cord 124 would be entered as “124s2s1.”